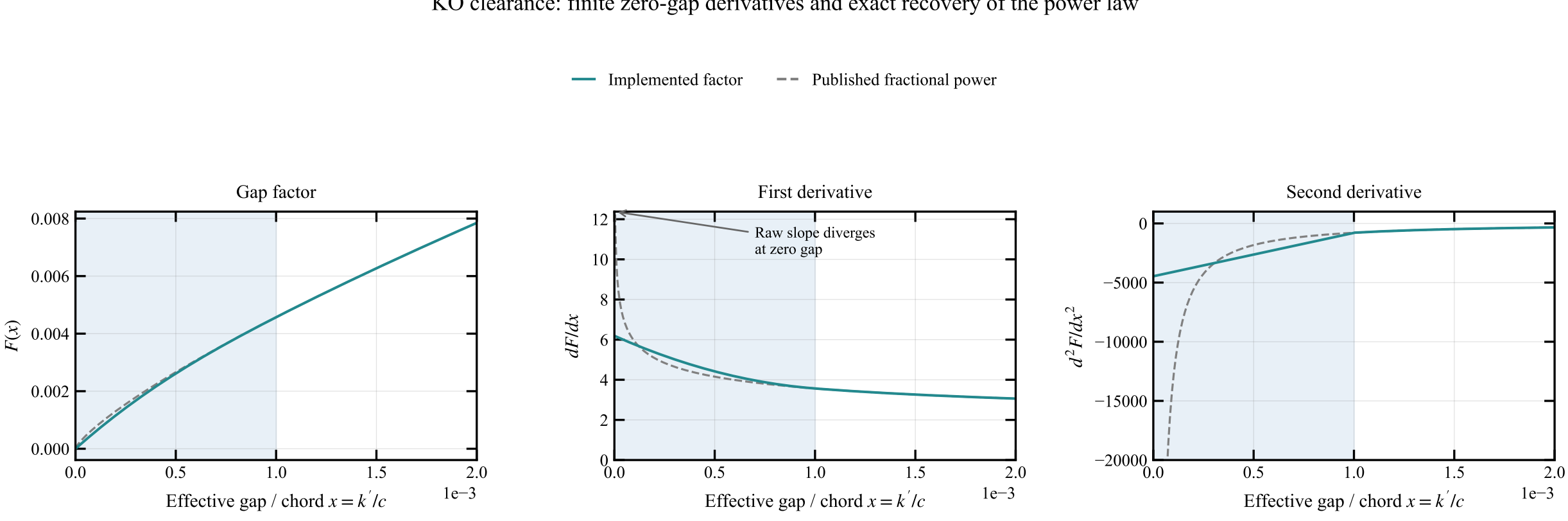

Blue band: numerical patch below effective gap/chord = 0.001; value, slope and curvature match at its upper edge.
The raw first and second derivatives are singular at zero and extend beyond the displayed limits.